

You are part of an organization that follows SRE practices and principles. You are taking over the management of a new service from the Development Team, and you conduct a Production Readiness Review (PRR). After the PRR analysis phase, you determine that the service cannot currently meet its Service Level Objectives (SLOs). You want to ensure that the service can meet its SLOs in production. What should you do next?

- A. Adjust the SLO targets to be achievable by the service so you can bring it into production.
 - B. Notify the development team that they will have to provide production support for the service.
 - C. Identify recommended reliability improvements to the service to be completed before handover. **Most Voted**
 - D. Bring the service into production with no SLOs and build them when you have collected operational data.
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You are running an experiment to see whether your users like a new feature of a web application. Shortly after deploying the feature as a canary release, you receive a spike in the number of 500 errors sent to users, and your monitoring reports show increased latency. You want to quickly minimize the negative impact on users. What should you do first?

- A. Roll back the experimental canary release. **Most Voted**
 - B. Start monitoring latency, traffic, errors, and saturation.
 - C. Record data for the postmortem document of the incident.
 - D. Trace the origin of 500 errors and the root cause of increased latency
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You are responsible for creating and modifying the Terraform templates that define your Infrastructure. Because two new engineers will also be working on the same code, you need to define a process and adopt a tool that will prevent you from overwriting each other's code. You also want to ensure that you capture all updates in the latest version. What should you do?

- A. ☒ Store your code in a Git-based version control system. ☒ Establish a process that allows developers to merge their own changes at the end of each day. ☒ Package and upload code to a versioned Cloud Storage basket as the latest master version.

- B. ☒☒☒ Store your code in a Git-based version control system. ☒☒☒ Establish a process that includes code reviews by peers and unit testing to ensure integrity and functionality before integration of code. ☒☒☒ Establish a process where the fully integrated code in the repository becomes the latest master version. **Most Voted**
 - C. ☒☒☒ Store your code as text files in Google Drive in a defined folder structure that organizes the files. ☒☒☒ At the end of each day, confirm that all changes have been captured in the files within the folder structure. ☒☒☒ Rename the folder structure with a predefined naming convention that increments the version.
 - D. ☒☒☒ Store your code as text files in Google Drive in a defined folder structure that organizes the files. ☒☒☒ At the end of each day, confirm that all changes have been captured in the files within the folder structure and create a new .zip archive with a predefined naming convention. ☒☒☒ Upload the .zip archive to a versioned Cloud Storage bucket and accept it as the latest version.
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You support a high-traffic web application with a microservice architecture. The home page of the application displays multiple widgets containing content such as the current weather, stock prices, and news headlines. The main serving thread makes a call to a dedicated microservice for each widget and then lays out the homepage for the user. The microservices occasionally fail; when that happens, the serving thread serves the homepage with some missing content. Users of the application are unhappy if this degraded mode occurs too frequently, but they would rather have some content served instead of no content at all. You want to set a Service Level Objective (SLO) to ensure that the user experience does not degrade too much. What Service Level Indicator (SLI) should you use to measure this?

- A. A quality SLI: the ratio of non-degraded responses to total responses. **Most Voted**
 - B. An availability SLI: the ratio of healthy microservices to the total number of microservices.
 - C. A freshness SLI: the proportion of widgets that have been updated within the last 10 minutes.
 - D. A latency SLI: the ratio of microservice calls that complete in under 100 ms to the total number of microservice calls.
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You support a multi-region web service running on Google Kubernetes Engine (GKE) behind a Global HTTP/S Cloud Load Balancer (CLB). For legacy reasons, user requests first go through a third-party Content Delivery Network (CDN), which then routes traffic to the CLB. You have already implemented an availability Service Level Indicator (SLI) at the CLB level. However, you want to increase coverage

in case of a potential load balancer misconfiguration, CDN failure, or other global networking catastrophe. Where should you measure this new SLI? (Choose two.)

- A. Your application servers' logs.
 - B. Instrumentation coded directly in the client. **Most Voted**
 - C. Metrics exported from the application servers.
 - D. GKE health checks for your application servers.
 - E. A synthetic client that periodically sends simulated user requests. **Most Voted**
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Your team is designing a new application for deployment into Google Kubernetes Engine (GKE). You need to set up monitoring to collect and aggregate various application-level metrics in a centralized location. You want to use Google Cloud Platform services while minimizing the amount of work required to set up monitoring. What should you do?

- A. Publish various metrics from the application directly to the Stackdriver Monitoring API, and then observe these custom metrics in Stackdriver. **Most Voted**
 - B. Install the Cloud Pub/Sub client libraries, push various metrics from the application to various topics, and then observe the aggregated metrics in Stackdriver.
 - C. Install the OpenTelemetry client libraries in the application, configure Stackdriver as the export destination for the metrics, and then observe the application's metrics in Stackdriver.
 - D. Emit all metrics in the form of application-specific log messages, pass these messages from the containers to the Stackdriver logging collector, and then observe metrics in Stackdriver.
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You support a production service that runs on a single Compute Engine instance. You regularly need to spend time on recreating the service by deleting the crashing instance and creating a new instance based on the relevant image. You want to reduce the time spent performing manual operations while following Site Reliability Engineering principles. What should you do?

- A. File a bug with the development team so they can find the root cause of the crashing instance.
- B. Create a Managed instance Group with a single instance and use health checks to determine the system status. **Most Voted**

- C. Add a Load Balancer in front of the Compute Engine instance and use health checks to determine the system status.
 - D. Create a Stackdriver Monitoring dashboard with SMS alerts to be able to start recreating the crashed instance promptly after it was crashed.
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Your application artifacts are being built and deployed via a CI/CD pipeline. You want the CI/CD pipeline to securely access application secrets. You also want to more easily rotate secrets in case of a security breach. What should you do?

- A. Prompt developers for secrets at build time. Instruct developers to not store secrets at rest.
 - B. Store secrets in a separate configuration file on Git. Provide select developers with access to the configuration file.
 - C. Store secrets in Cloud Storage encrypted with a key from Cloud KMS. Provide the CI/CD pipeline with access to Cloud KMS via IAM. **Most Voted**
 - D. Encrypt the secrets and store them in the source code repository. Store a decryption key in a separate repository and grant your pipeline access to it.
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Your company follows Site Reliability Engineering practices. You are the person in charge of Communications for a large, ongoing incident affecting your customer-facing applications. There is still no estimated time for a resolution of the outage. You are receiving emails from internal stakeholders who want updates on the outage, as well as emails from customers who want to know what is happening. You want to efficiently provide updates to everyone affected by the outage. What should you do?

- A. Focus on responding to internal stakeholders at least every 30 minutes. Commit to a "next update" time.
 - B. Provide periodic updates to all stakeholders in a timely manner. Commit to a "next update" time in all communications. **Most Voted**
 - C. Delegate the responding to internal stakeholder emails to another member of the Incident Response Team. Focus on providing responses directly to customers.
 - D. Provide all internal stakeholder emails to the Incident Commander, and allow them to manage internal communications. Focus on providing responses directly to customers.
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Your team uses Cloud Build for all CI/CD pipelines. You want to use the kubectl builder for Cloud Build to deploy new images to Google Kubernetes Engine (GKE). You need to authenticate to GKE while minimizing development effort. What should you do?

- A. Assign the Container Developer role to the Cloud Build service account. **Most Voted**
 - B. Specify the Container Developer role for Cloud Build in the cloudbuild.yaml file.
 - C. Create a new service account with the Container Developer role and use it to run Cloud Build.
 - D. Create a separate step in Cloud Build to retrieve service account credentials and pass these to kubectl.
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You support an application that stores product information in cached memory. For every cache miss, an entry is logged in Stackdriver Logging. You want to visualize how often a cache miss happens over time. What should you do?

- A. Link Stackdriver Logging as a source in Google Data Studio. Filter the logs on the cache misses.
 - B. Configure Stackdriver Profiler to identify and visualize when the cache misses occur based on the logs.
 - C. Create a logs-based metric in Stackdriver Logging and a dashboard for that metric in Stackdriver Monitoring. **Most Voted**
 - D. Configure BigQuery as a sink for Stackdriver Logging. Create a scheduled query to filter the cache miss logs and write them to a separate table.
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You need to deploy a new service to production. The service needs to automatically scale using a Managed Instance Group (MIG) and should be deployed over multiple regions. The service needs a large number of resources for each instance and you need to plan for capacity. What should you do?

- A. Use the n1-highcpu-96 machine type in the configuration of the MIG.
- B. Monitor results of Stackdriver Trace to determine the required amount of resources.
- C. Validate that the resource requirements are within the available quota limits of each region. **Most Voted**

- D. Deploy the service in one region and use a global load balancer to route traffic to this region.
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You are running an application on Compute Engine and collecting logs through Stackdriver. You discover that some personally identifiable information (PII) is leaking into certain log entry fields. All PII entries begin with the text `userinfo`. You want to capture these log entries in a secure location for later review and prevent them from leaking to Stackdriver Logging. What should you do?

- A. Create a basic log filter matching `userinfo`, and then configure a log export in the Stackdriver console with Cloud Storage as a sink.
 - B. Use a Fluentd filter plugin with the Stackdriver Agent to remove log entries containing `userinfo`, and then copy the entries to a Cloud Storage bucket. **Most Voted**
 - C. Create an advanced log filter matching `userinfo`, configure a log export in the Stackdriver console with Cloud Storage as a sink, and then configure a log exclusion with `userinfo` as a filter.
 - D. Use a Fluentd filter plugin with the Stackdriver Agent to remove log entries containing `userinfo`, create an advanced log filter matching `userinfo`, and then configure a log export in the Stackdriver console with Cloud Storage as a sink.
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You have a CI/CD pipeline that uses Cloud Build to build new Docker images and push them to Docker Hub. You use Git for code versioning. After making a change in the Cloud Build YAML configuration, you notice that no new artifacts are being built by the pipeline. You need to resolve the issue following

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Reliability Engineering practices. What should you do?

- A. Disable the CI pipeline and revert to manually building and pushing the artifacts.
 - B. Change the CI pipeline to push the artifacts to Container Registry instead of Docker Hub.
 - C. Upload the configuration YAML file to Cloud Storage and use Error Reporting to identify and fix the issue.
 - D. Run a Git compare between the previous and current Cloud Build Configuration files to find and fix the bug. **Most Voted**
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Your company follows Site Reliability Engineering principles. You are writing a postmortem for an incident, triggered by a software change, that severely affected users. You want to prevent severe incidents from happening in the future. What should you do?

- A. Identify engineers responsible for the incident and escalate to their senior management.
 - B. Ensure that test cases that catch errors of this type are run successfully before new software releases. **Most Voted**
 - C. Follow up with the employees who reviewed the changes and prescribe practices they should follow in the future.
 - D. Design a policy that will require on-call teams to immediately call engineers and management to discuss a plan of action if an incident occurs.
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You support a high-traffic web application that runs on Google Cloud Platform (GCP). You need to measure application reliability from a user perspective without making any engineering changes to it. What should you do? (Choose two.)

- A. Review current application metrics and add new ones as needed.
 - B. Modify the code to capture additional information for user interaction.
 - C. Analyze the web proxy logs only and capture response time of each request.
 - D. Create new synthetic clients to simulate a user journey using the application. **Most Voted**
 - E. Use current and historic Request Logs to trace customer interaction with the application. **Most Voted**
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You manage an application that is writing logs to Stackdriver Logging. You need to give some team members the ability to export logs. What should you do?

- A. Grant the team members the IAM role of logging.configWriter on Cloud IAM. **Most Voted**
 - B. Configure Access Context Manager to allow only these members to export logs.
 - C. Create and grant a custom IAM role with the permissions logging.sinks.list and logging.sink.get.
 - D. Create an Organizational Policy in Cloud IAM to allow only these members to create log exports.
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Your application services run in Google Kubernetes Engine (GKE). You want to make sure that only images from your centrally-managed Google Container Registry (GCR) image registry in the altostrat-images project can be deployed to the cluster while minimizing development time. What should you do?

- A. Create a custom builder for Cloud Build that will only push images to gcr.io/altostrat-images.
 - B. Use a Binary Authorization policy that includes the whitelist name pattern gcr.io/altostrat-images/. **Most Voted**
 - C. Add logic to the deployment pipeline to check that all manifests contain only images from gcr.io/altostrat-images.
 - D. Add a tag to each image in gcr.io/altostrat-images and check that this tag is present when the image is deployed.
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Your team has recently deployed an NGINX-based application into Google Kubernetes Engine (GKE) and has exposed it to the public via an HTTP Google Cloud Load Balancer (GCLB) ingress. You want to scale the deployment of the application's frontend using an appropriate Service Level Indicator (SLI). What should you do?

- A. Configure the horizontal pod autoscaler to use the average response time from the Liveness and Readiness probes.
 - B. Configure the vertical pod autoscaler in GKE and enable the cluster autoscaler to scale the cluster as pods expand.
 - C. Install the Stackdriver custom metrics adapter and configure a horizontal pod autoscaler to use the number of requests provided by the GCLB. **Most Voted**
 - D. Expose the NGINX stats endpoint and configure the horizontal pod autoscaler to use the request metrics exposed by the NGINX deployment.
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Your company follows Site Reliability Engineering practices. You are the Incident Commander for a new, customer-impacting incident. You need to immediately assign two incident management roles to assist you in an effective incident response. What roles should you assign? (Choose two.)

- A. Operations Lead **Most Voted**

- B. Engineering Lead
 - C. Communications Lead **Most Voted**
 - D. Customer Impact Assessor
- E. External Customer Communications Lead
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You support an application running on GCP and want to configure SMS notifications to your team for the most critical alerts in Stackdriver Monitoring. You have already identified the alerting policies you want to configure this for. What should you do?

- A. Download and configure a third-party integration between Stackdriver Monitoring and an SMS gateway. Ensure that your team members add their SMS/phone numbers to the external tool.
 - B. Select the Webhook notifications option for each alerting policy, and configure it to use a third-party integration tool. Ensure that your team members add their SMS/phone numbers to the external tool.
 - C. Ensure that your team members set their SMS/phone numbers in their Stackdriver Profile. Select the SMS notification option for each alerting policy and then select the appropriate SMS/phone numbers from the list. **Most Voted**
- D. Configure a Slack notification for each alerting policy. Set up a Slack-to-SMS integration to send SMS messages when Slack messages are received. Ensure that your team members add their SMS/phone numbers to the external integration.
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You are managing an application that exposes an HTTP endpoint without using a load balancer. The latency of the HTTP responses is important for the user experience. You want to understand what HTTP latencies all of your users are experiencing. You use Stackdriver Monitoring. What should you do?

- A. ⌘ In your application, create a metric with a metricKind set to DELTA and a valueType set to DOUBLE. ⌘ In Stackdriver's Metrics Explorer, use a Stacked Bar graph to visualize the metric.
- B. ⌘ In your application, create a metric with a metricKind set to CUMULATIVE and a valueType set to DOUBLE. ⌘ In Stackdriver's Metrics Explorer, use a Line graph to visualize the metric.

- C. ☐ In your application, create a metric with a metricKind set to GAUGE and a valueType set to DISTRIBUTION. ☐ In Stackdriver's Metrics Explorer, use a Heatmap graph to visualize the metric. **Most Voted**
 - D. ☐ In your application, create a metric with a metricKind set to METRIC_KIND_UNSPECIFIED and a valueType set to INT64. ☐ In Stackdriver's Metrics Explorer, use a Stacked Area graph to visualize the metric.
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Your team is designing a new application for deployment both inside and outside Google Cloud Platform (GCP). You need to collect detailed metrics such as system resource utilization. You want to use centralized GCP services while minimizing the amount of work required to set up this collection system. What should you do?

- A. Import the Stackdriver Profiler package, and configure it to relay function timing data to Stackdriver for further analysis. **Most Voted**
 - B. Import the Stackdriver Debugger package, and configure the application to emit debug messages with timing information.
 - C. Instrument the code using a timing library, and publish the metrics via a health check endpoint that is scraped by Stackdriver.
 - D. Install an Application Performance Monitoring (APM) tool in both locations, and configure an export to a central data storage location for analysis.
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You need to reduce the cost of virtual machines (VM) for your organization. After reviewing different options, you decide to leverage preemptible VM instances. Which application is suitable for preemptible VMs?

- A. A scalable in-memory caching system. **Most Voted**
 - B. The organization's public-facing website.
 - C. A distributed, eventually consistent NoSQL database cluster with sufficient quorum.
 - D. A GPU-accelerated video rendering platform that retrieves and stores videos in a storage bucket.
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Your organization recently adopted a container-based workflow for application development. Your team develops numerous applications that are deployed continuously through an automated build pipeline to a Kubernetes cluster in the production environment. The security auditor is concerned that developers or operators could circumvent automated testing and push code changes to production without approval. What should you do to enforce approvals?

- A. Configure the build system with protected branches that require pull request approval.
 - B. Use an Admission Controller to verify that incoming requests originate from approved sources.
 - C. Leverage Kubernetes Role-Based Access Control (RBAC) to restrict access to only approved users.
 - D. Enable binary authorization inside the Kubernetes cluster and configure the build pipeline as an attestor. **Most Voted**
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